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How can swimming be less of a drag?

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Abstract

It is important for animals to move around easily. It helps them avoid predators or catch a meal. But the ways animals move depend on the environment they live in. A lot of animals that live in water beat their legs one after another. It is called metachronal swimming. And it is really efficient at propelling animals through the water. We wanted to investigate the other half of the story. Is metachronal swimming also good at reducing drag?

We performed experiments on grass shrimp to look at metachronal swimming. We looked at leg flexibility, the

patterns of leg movement, and how the water moved around the legs. Then we built a robotic model to measure water flow and force data at the same time. We found that leg flexibility reduced drag up to 75% compared to stiff legs. We also saw that there was less drag when legs grouped together. It caused fewer interactions with wakes from other legs. Hopefully, this information can help us build more efficient underwater vehicles in the future.

Introduction

Have you ever tried to walk through a swimming pool or a lake? It can be hard. Small animals like shrimp and krill can have a hard time moving around in the water. The water can seem sticky to the small animals that live in it, causing a lot of friction. Luckily, many animals have traits that can help them move more easily through water.

Some animals perform **metachronal swimming**. This is when an animal beats their legs one after another through the water. Sometimes the legs even come together and act as a group. It is a coordinated movement that produces forward **thrust**. And it works better than if the legs beat on their own.

Leg flexibility also helps animals move in the water. If a leg is stiff on the **power stroke**, it can generate a lot of thrust. But if it is stiff during the **recovery stroke** (getting back to the beginning), it can produce a lot of **drag**. There is less resistance from the water if the leg bends. Think about moving through a waist-deep pool. If you keep your legs straight, you can't move very fast. And it takes a lot of energy. But if you bend your knees to walk through the water, you can move more easily.

Researchers know a lot about how swimming style and leg flexibility can increase thrust. But we don't know a lot about how they reduce drag. We had several questions. **How much does leg flexibility contribute to reduced drag? Does metachronal swimming reduce drag?** By how much? This type of information could help us design new underwater robots in the future.



Grass shrimp are also variously known as the common American prawn, marsh grass shrimp or marsh shrimp.

Methods

→ Leg Flexibility

We took legs from recently deceased grass shrimp. We scanned the legs to figure out how curved they were. Then we mounted the legs to a support that didn't move. We used a pin to push on each side of the leg. Then we calculated how much the leg bent in each direction.

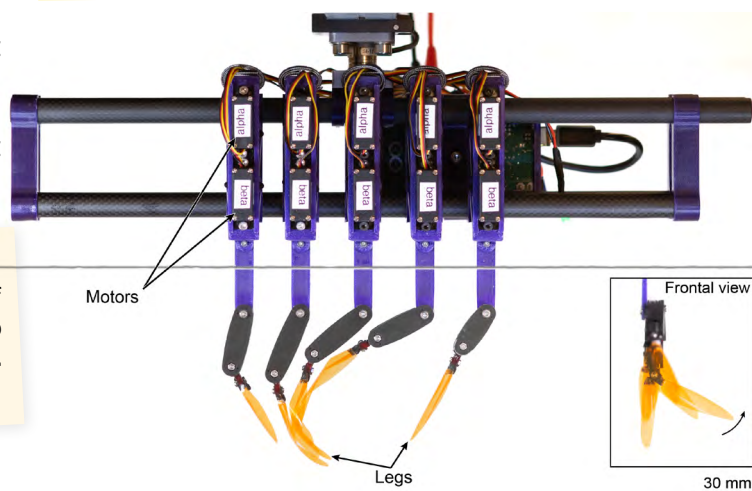
→ Metachronal Swimming

We looked at live grass shrimp in an aquarium. We observed their recovery stroke. We timed how long each leg was coordinated with or touching the other legs. We compared this to the length of the recovery stroke. We also looked at water flow around the shrimp's legs. We did this by putting very small particles in the water. Then we illuminated the particles with a thin sheet of laser light. We used a very fast camera to see how the particles moved around the legs.

Figure 1: Our robotic shrimp model was based on measurements of live grass shrimp. The legs were flexible in one direction. We could also control individual legs to make the robot swim in different patterns. Notice how the legs group together as it moves.

→ Robotic Shrimp

We used our measurements from live animals to design a robotic shrimp (Fig. 1). We made it using 3D printing, plastic heat shaping, and electronics. Our robot was 20 times bigger than a real grass shrimp. We put our robot in a fluid that mimicked the conditions that our live grass shrimp lived in. We looked at the robot swimming with different stroke patterns. We also looked at water flow around the legs. Finally we calculated real-time thrust and drag measurements using sensors on the robot.



Results

1. We saw that the back face of the grass shrimp's legs was curved. The legs were up to 1.7 times stiffer when we pushed on the back (similar to the power stroke) compared to the front (similar to the recovery stroke). We found that this reduced drag in individual robotic legs by up to 75%.

2. In the aquarium, we observed the middle three legs (out of 5) come together and move as a group. In our robot model, we found that metachronal swimming increased total thrust by about 30%. This was compared to independent leg swimming (Fig. 2).

Did metachronal swimming increase or decrease net force relative to independently moving legs? By how much?

3. We saw that water flow around the shrimp's legs created vortices. These are the small wakes the legs create as they move through the water. Vortices can increase drag if the other legs try to swim through them. We found that metachronal swimming had the fewest interactions with vortices.

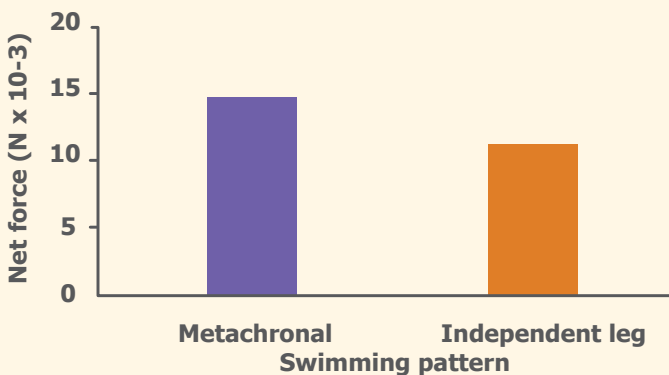


Figure 2: The amount of net force generated for our robot during metachronal versus independent leg swimming patterns. Net force is the sum of thrust and drag. If drag is greater then the net force will be less.

Discussion

Our results suggest that moving through water isn't just about thrust. It's also about drag. Bending can reduce the area of the leg moving through water. This reduces drag during the recovery stroke. Bending also allows the legs to group together. The first leg in a group acts as a shield, protecting the legs behind it from the effects of water. This then reduces drag and there are fewer vortex interactions.

Reducing drag helps animals move faster and further through the water. They might be able to escape from

predators better. It could also help them catch food. Shrimp, **comb jellies**, bristle worms, and **copepods** are all metachronal swimmers. It is a common way to swim. This suggests it **evolved** many times because it was so helpful.

We should consider metachronal swimming when we design underwater vehicles. Many designs focus on maximizing thrust. What if we could reduce drag instead? Could we make underwater vehicles that use less energy? Hopefully, we will find out!

Conclusion

Our research shows that movement can be complex. And it is specific to the environment an animal lives in. These traits have evolved over millions of years. And we are just learning how some of them work!

The next time you go to the zoo or a park, take a look at all the different ways animals move. Do they swim? Fly?

Or maybe they slither along the ground? Then look at the traits that allow them to move that way. Do they have wings, or a tail, or sticky skin? Can you figure out what advantages those traits give the animal? How do they help it move and survive?

Glossary of Key Terms

3D printing - a manufacturing process that creates three-dimensional objects from a digital design by adding material layer by layer. Unlike traditional manufacturing methods that remove material to create a shape, 3D printing builds objects by placing material only where it is needed.

Comb jellies - marine organisms without a backbone. They have rows of tiny hairs that they use to swim.

Copepods - small crustaceans that live in both fresh and saltwater and have multiple legs to help them swim.

Drag - a force acting in the opposite direction of movement. The drag depends on how much friction an object feels and how fast the object is moving. For example, less drag will act on a plane in the air than in the water (if it is going the same speed).

Evolution - change over time guided by natural selection. Traits that enhance survival and reproduction get passed on to the next generation. Over time these traits become more common in the population.

Metachronal swimming - a swimming technique where multiple legs stroke back and forth sequentially. The movement is coordinated and sometimes legs even group together. It is seen in a variety of animals that live in the water.

Power stroke - the active part of swimming that generates force forward (thrust). This is usually a leg or paddle-like structure that pulls or pushes the animal through the water.

Recovery stroke - the part of swimming that returns the leg to the beginning position.

Thrust - a force that moves an organism forward.

Vortex - a region where water rotates. You can create vortices in water just by dragging your hand through it.

Wake - an area of disturbed and rotating water behind an organism. This can be caused by the water moving around the organism or the organism moving through the water.

REFERENCES

Nils B. Tack, Sara O. Santos, and Monica M. Wilhelmus (2025) *Going around the bend to understand the role of leg coalescence in metachronal swimming*. Journal of Experimental Biology.

<https://journals.biologists.com/jeb/article/228/12/jeb249330/368328/Going-around-the-bend-to-understand-the-role-of>

A-Z Animals: Swimming

<https://a-z-animals.com/locomotion/swimming/>

Kiddle: Drag (physics) facts for kids

[https://kids.kiddle.co/Drag_\(physics\)](https://kids.kiddle.co/Drag_(physics))

NASA: What is aerodynamics?

<https://www.nasa.gov/learning-resources/for-kids-and-students/what-is-aerodynamics-grades-k-4/>

Check your understanding

- 1 What is the difference between drag and thrust? If thrust is greater than drag, will the animal move?
- 2 Why did we decide to build a robot? What important information did we get from a robot that we couldn't get from live animals?
- 3 Why does metachronal swimming with legs that bend reduce drag?
- 4 Think about your own life. Describe three situations where drag plays a factor in your daily activities. Suggest one way for each situation that you could reduce drag if you wanted to.
- 5 Do some research online. Pick an underwater vehicle that exists. It could be a submarine, an autonomous underwater vehicle, or some kind of remotely operated underwater vehicle. Make three suggestions for changing the design of your underwater vehicle that would reduce drag.

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